Zero to Understanding with Oracle as a Microsoft Pro

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She/Her

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I’m the AI and Multi-Platform Advocate at Redgate Software. Along with presenting on open source, AI and Oracle products, I also work with the engineering and product teams on Redgate solutions.

I’ve been in the tech industry for over 27 years and am known online under my handle, DBAKevlar.

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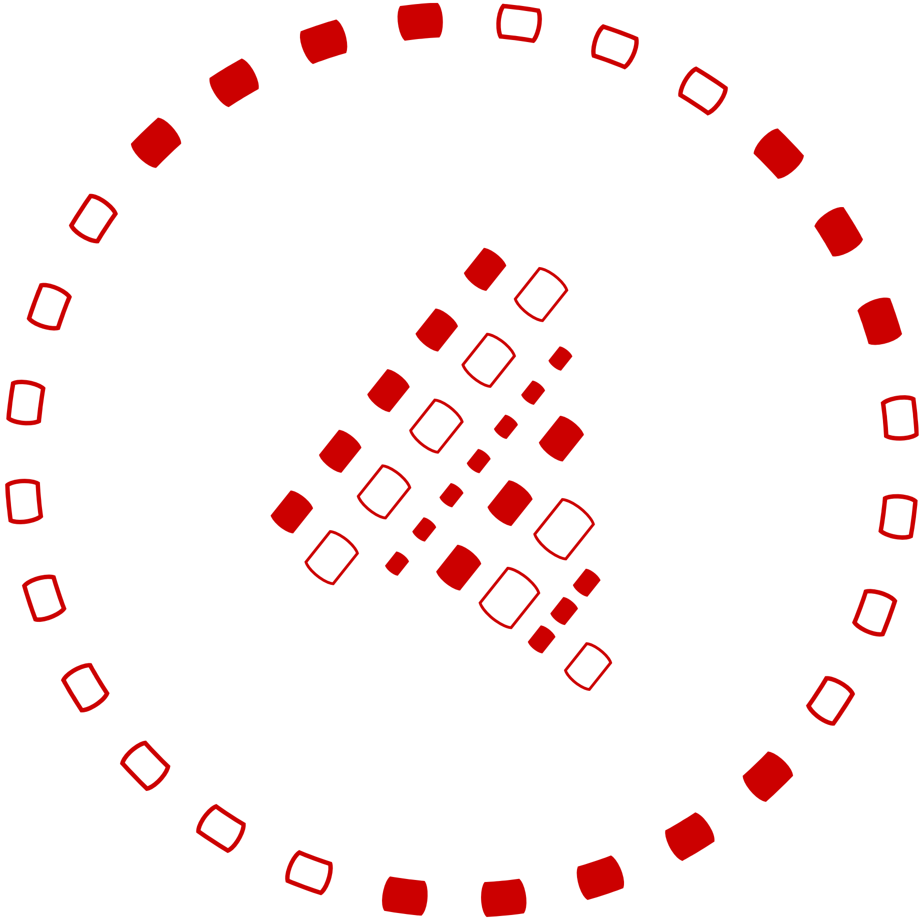
I’m an innovative Principal Cloud Solution Architect with extensive experience in designing and implementing end-to-end technology projects across multiple industries.

I focus on success in leading cross-functional teams, fostering collaboration to ensure seamless execution of complex cloud architectures. Community-driven and dynamic presenter, with extensive international footprint, recognized for delivering insightful perspectives to diverse audiences in 30+ countries.

Breakfast Agenda

- Conversation
- Interactive
- Value Positive

Oracle Peanut Butter in my SQL Server Chocolate



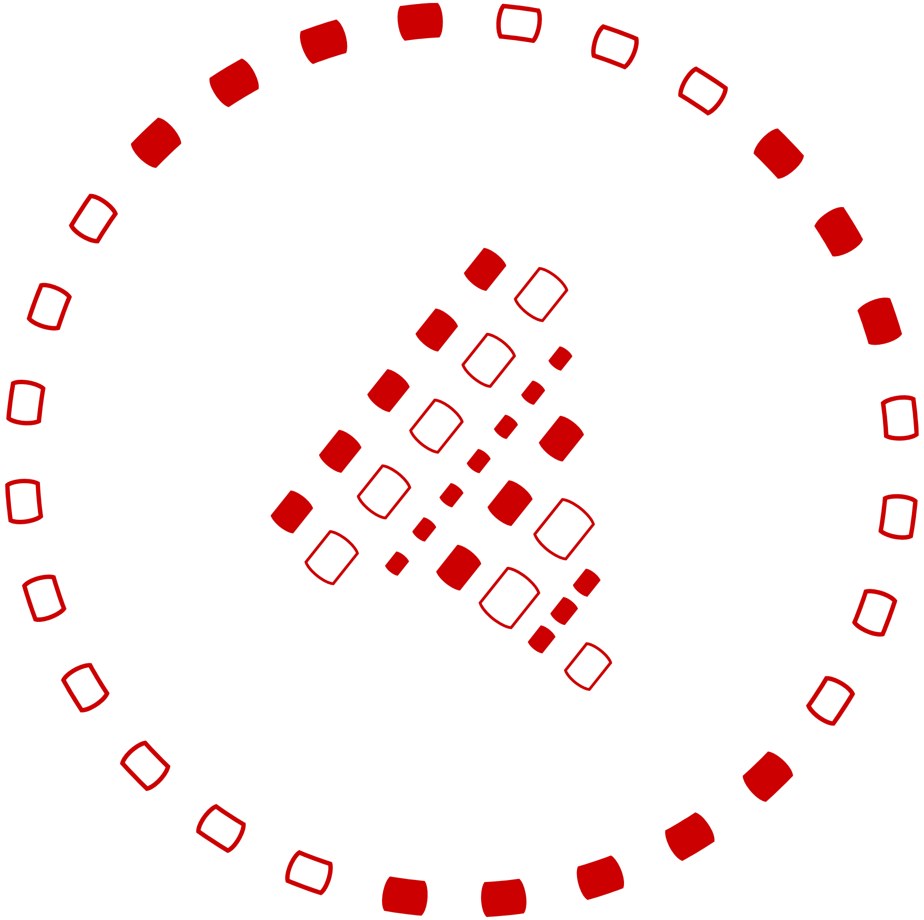
A Quick Poll

- Experience Level with Oracle?
  - New to Oracle
  - 1-5 years
  - 5-8 years
  - 8+ years

Daily Focus Poll

- My daily focus is…
  - Development
  - Production
  - End User
  - Leadership

The Oracle Ecosystem



Oracle – Key Facts

📌 **Founded** – 1977 by Larry Ellison, Bob Miner, and Ed Oates
📌 **Headquarters** – Austin, Texas, but soon to be - Nashville, Tennessee
📌 **Global Presence** – Operates in 175+ countries
📌 **Revenue** – $50B+ annually
📌 **Employees** – 140,000+ worldwide
📌 **Major Customers** – Fortune 500 companies, governments, and enterprises
📌 **Innovation Focus** – AI-driven automation, multi-cloud, security

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Oracle, is Everywhere

Database, software, hardware, cloud

Fortune 500 enterprise and large organizations

Mostly on-premises, but more running in the cloud

Database environments can be TB or PB in size!

Oracle Database Versioning

- No Man's Land: **Oracle 8, 8i, 9i, 10g and 11g**
- Still Holding on: **Oracle 12.1 and 12.2**(extended support)
- Non-terminal Release, but supported: **Oracle 18c, 21c** (AWS RDS)
- Oracle Terminal Release, Full Supported: **Oracle 19c**
  - Exceptionally buggy version!
  - Should run latest release.
  - Best on Oracle Enterprise Linux, (OEL or EL, followed by Red Hat)
- **Oracle 23ai:** Bleeding Edge, but GA!!

"I thought Oracle was a Database Platform??"

Hardware Vendor?

Application Suites?

Storage Management System?

Cloud Vendor?

Coding Language?

Oracle Hardware

- Oracle purchased Sun Microsystems in 2010.
  - Zettabyte File System(ZFS) for inexpensive storage solution
  - Zero Data Loss Recovery Appliance (ZDLRA) for backup and recovery
  - Oracle Data Appliance (ODA) aka "Poor man's Exadata)

*We'll cover Exadata separately as it is an engineered system.

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Oracle Application Suites

E-business Suite- Accounts Receivable, Order Entry, Customer Orders, etc. Multi-tier system with Concurrent Manager for job management.

Peoplesoft- HR software suite

Hyperion/EPM – Financial and Operation performance Software

JD Edwards – Financial System

Primavera – Project Management Software

Opera – Hotel and Resort Management

APEX – Application Express**,** Web application creation suite

Storage Management

ASM Diskgroups

- ASM Diskgroups at the end of the report period for the database

Disk Group	Size (GB)	Used (GB)	# Griddisks	Redundancy	State
DATAC1	1,663,944.00	562,243.49	228	NORMAL	CONNECTED
RECOC1	416,142.75	112,022.39	228	NORMAL	CONNECTED

Automatic Storage Management, ASM

- Like NFTS or FAT32 in Windows
- Very similar to Linux Volume Manager(LVM) and UDEV in Linux/Unix

Possesses its own Oracle instance that is running and managing the storage layer for Oracle.

Removes the secondary layer at the OS level and is optimized for Oracle databases

Has its own syntax, utilities, views and command line reference.

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Oracle and the Cloud Vendor

- Oracle Cloud Infrastructure (OCI)
  - This is Oracle's 5th public cloud offering.
  - PaaS and SaaS offerings built out of 4th cloud, Oracle Bare Metal.
  - Free tier of Oracle cloud is 3rd public cloud "repurposed".

  - Made up mostly of Oracle Data Appliances (for SaaS offerings) and significantly now offers Oracle Exadata solutions with the PaaS database.
  - Oracle's PaaS database is called the Autonomous Database or Oracle Autonomous DB, (ADB).
  - Exadata@Cloud are Exadata in other cloud datacenters (Azure and GCP, recently in AWS.)

Oracle as Software

- PL/SQL – Flavor of SQL with incredibly robust packages to simplify development work. Can be found in Oracle, DB2 and extension for PostgreSQL.
  - Want the most from Oracle? Use PL/SQL, not "vanilla" SQL.
- Java – Java is one of the most popular coding languages in the world and incredibly robust. Oracle receives royalties off of most Java implementations in vendor code.
- Oracle also owns the trademark for Javascript.

Real Oracle Licensing Auditors

Oracle License Contracts are Unique

- Oracle licensing costs are commonly the most expensive part of any database project.
- Oracle is licensed by CORE, management pack.
- There are unlimited licensing agreements (EULAs) with large customers.
- Customer’s rarely pay list price- often large discounts in end.
- Annual support cost on top of licensing.

Oracle Specialists Are Focused

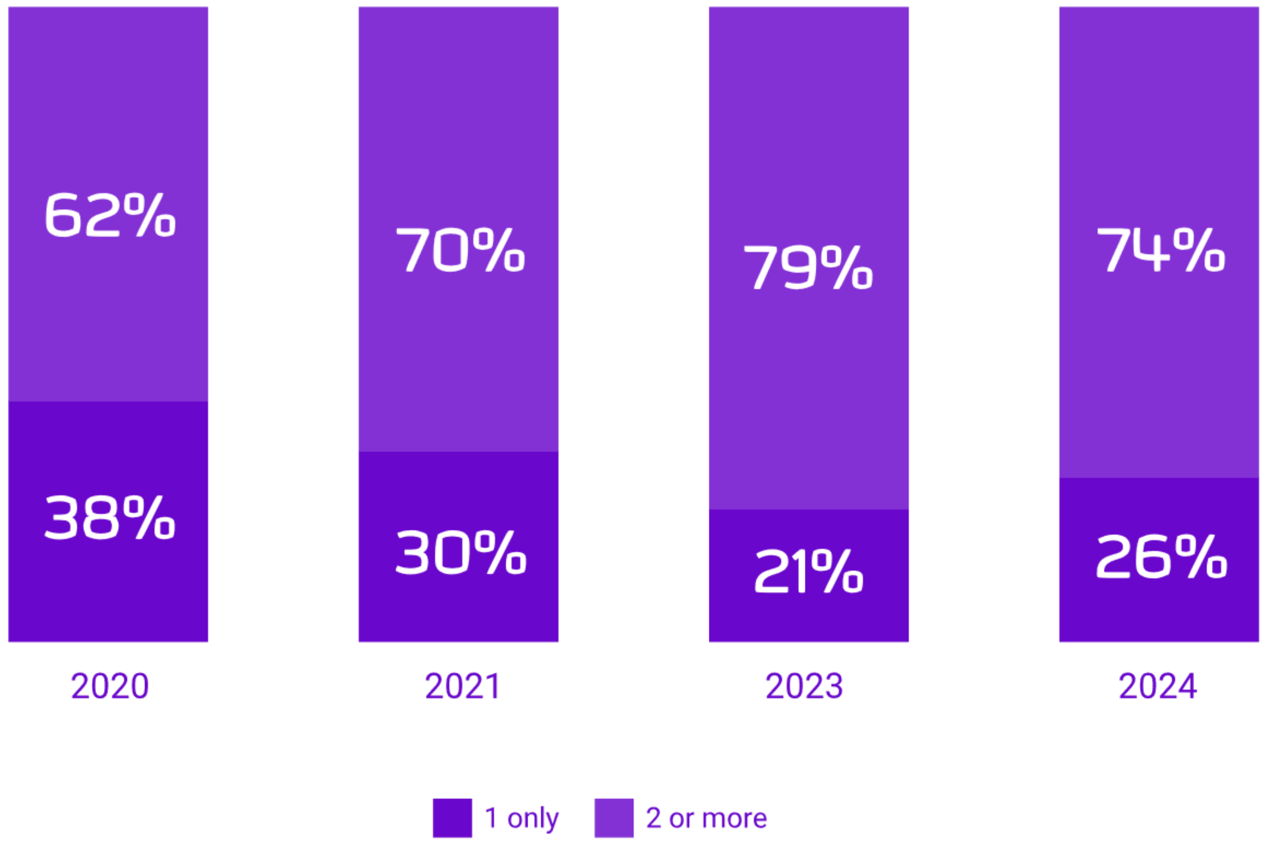
Database Administrator

- Development DBA
- Production DBA
- Application DBA

Developer

- PL/SQL Developer
- Application Developer
- Web Developer

Multi-platform is the norm



NEW Redgate State of Database Landscape Report 2025

After 27 years, I still have challenges "switching"

Same terms, different definitions

Different tools

Different uses for different platforms

Little understanding, "A database is a database…"

Increasing data volume and platform type

74%

of IT teams are now using more than one database platform

25%

of respondents using more than four database platforms

NEW Redgate State of Database Landscape Report 2025

The Oracle Database

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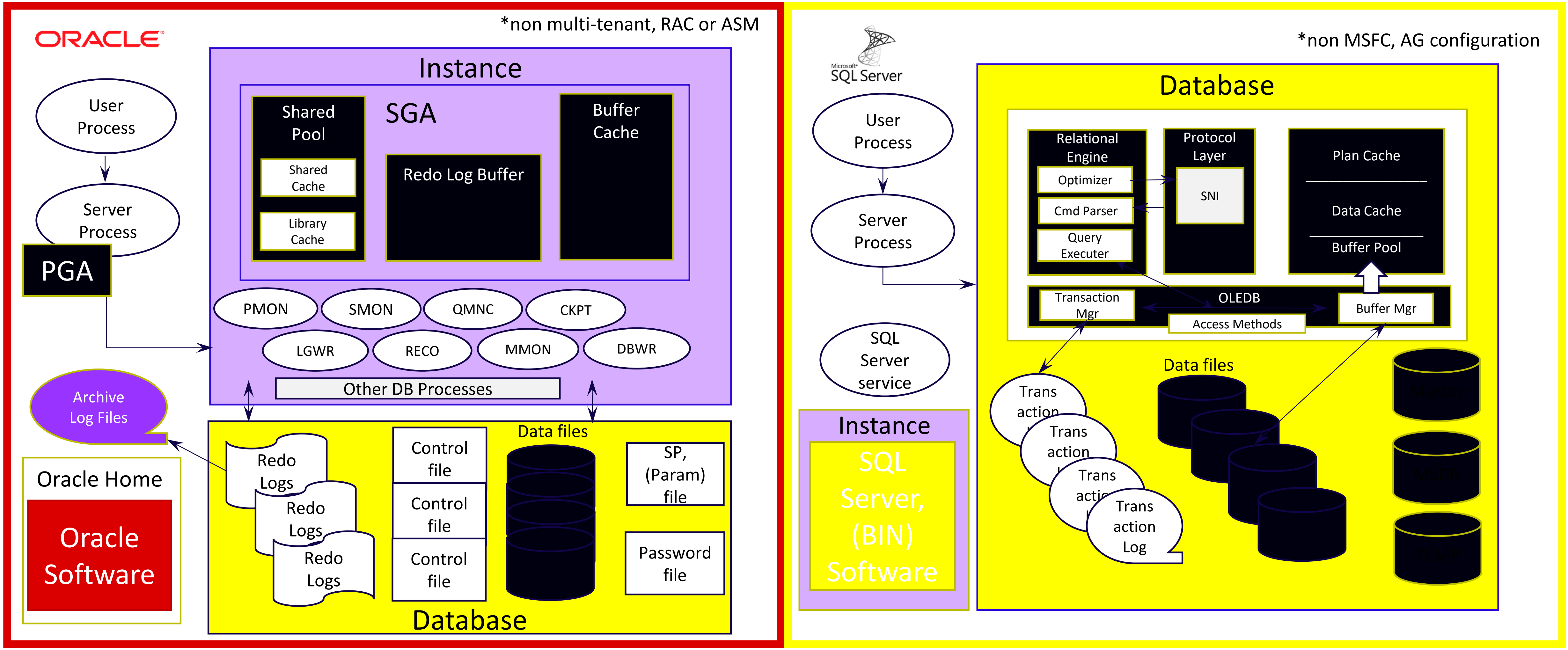
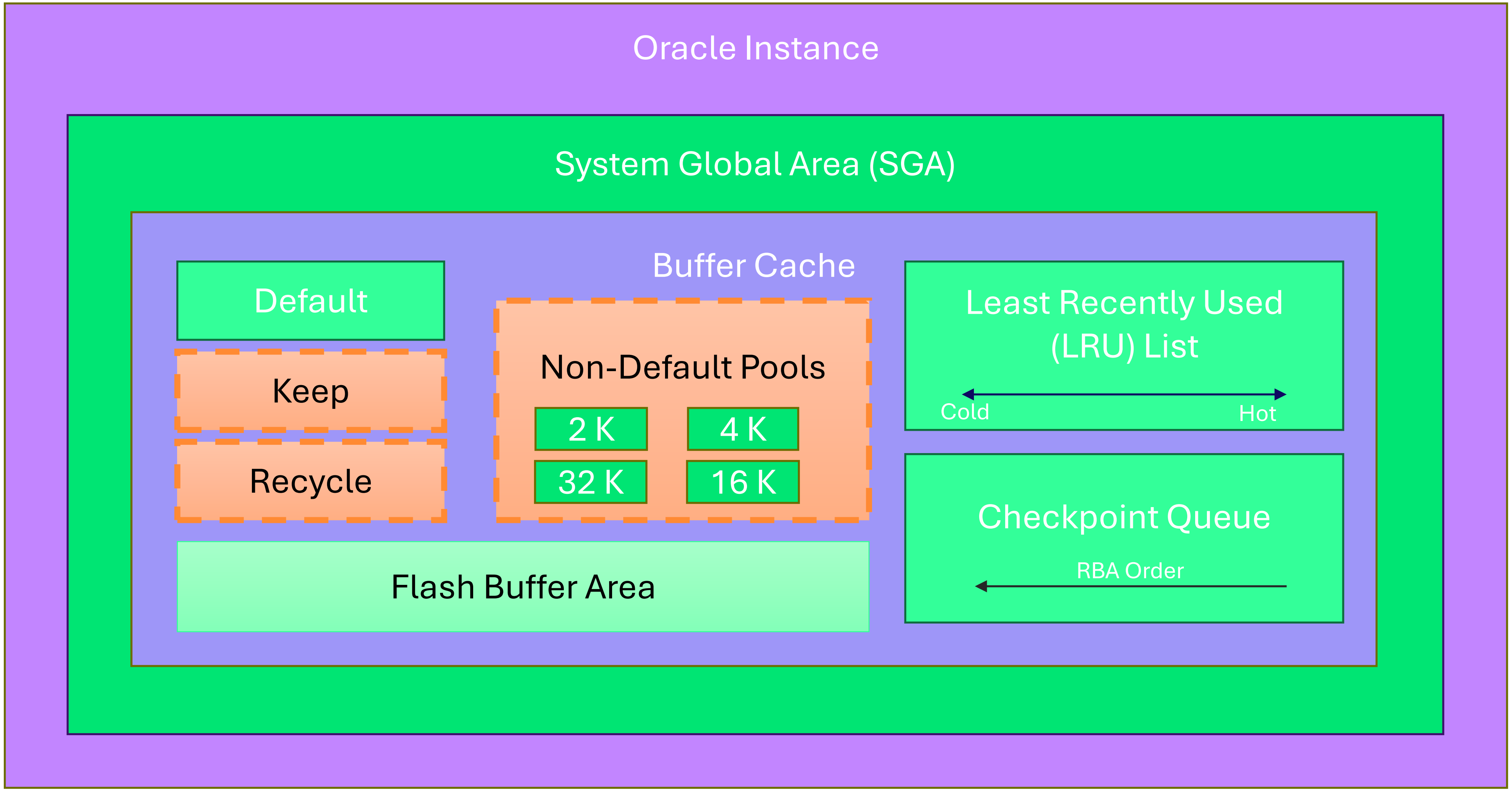
Terms to Clarify

- Instance
- Database
- Locking
- Buckets
- Process/Session
- Logs, (redo, archive, transaction)
- Schema/DBO
- Role/Group
- Cluster
- Packages
- Before Triggers/Complex Rules
- Synonyms
- Sequences/Identity Columns

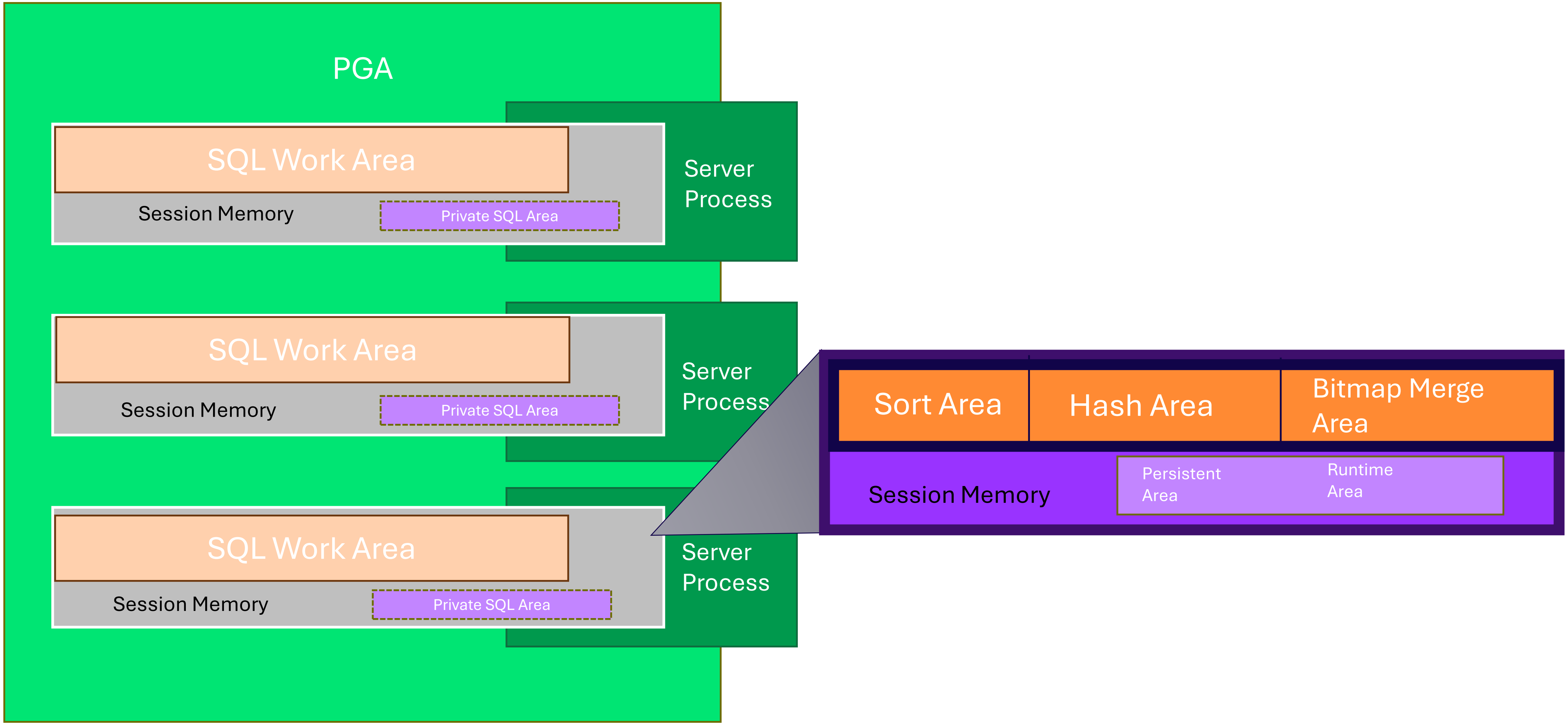
Oracle Compared to SQL Server, (Database Level)



Memory: Oracle SGA



Memory: Oracle PGA



What is an Oracle Instance?

- Background processes and memory allocation
- From SQLPlus, SQLcl or other utility-

`export|set ORACLE_SID=ORCL, ORACLE_HOME, ORACLE_BASE`

```
>sqlplus / as sysdba
```

Staged opening of an Oracle Database:

```
>startup nomount|mount|open
```

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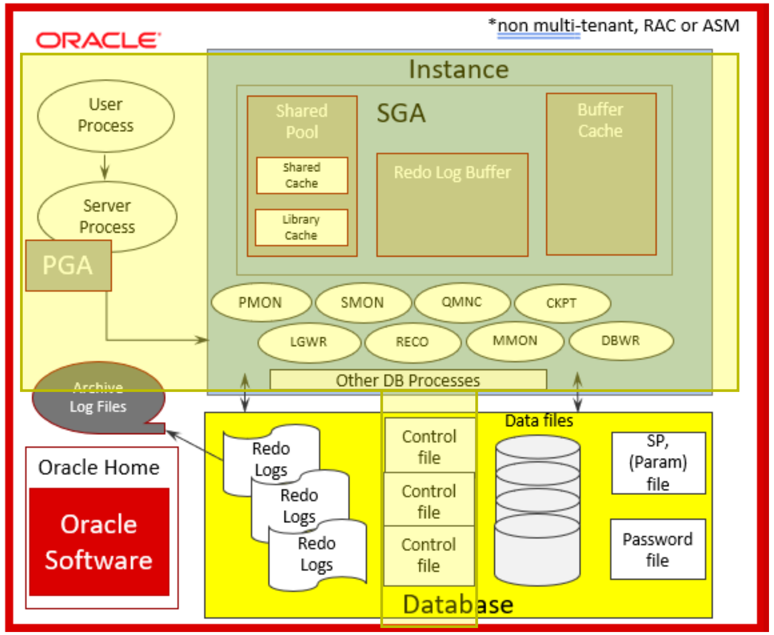
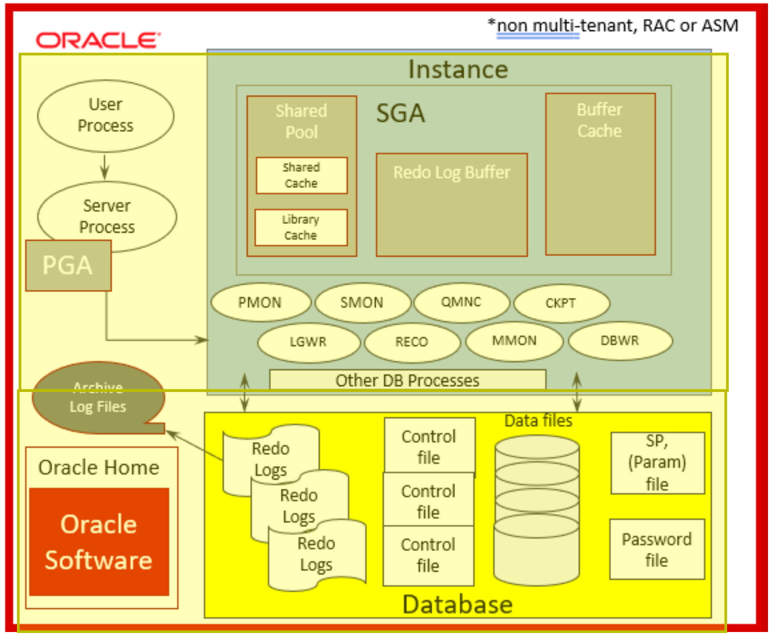
Aspects of the Startup

- NoMount



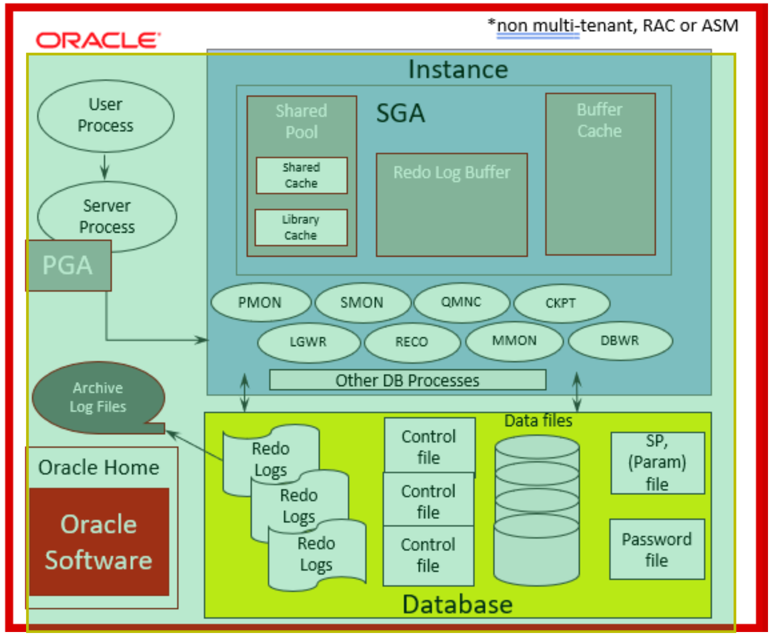
Memory and processes open, control files and parameters initialized.

- Mount



Everything from NOMOUNT and all datafiles, redo logs, etc. are mounted, but the database is NOT open

- Open



The database is not open after performing all previous steps.

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Network Connectivity

- Listener Service/Utility listens for connections
  - Listener.ora file contains listener configuration
  - Tnsnames.ora contains service names connections
  - Sqlnet.ora contains network configuration information

Users, Schemas and Object Storage

- User databases are broken up by schemas
- Each schema will have objects isolated by their own logical tablespaces, made up of 1 or more datafiles, (common).

Storage

- Byte→ Block→ Extent→ Segment→ Tablespace→ Database
- Every datafile(s) belong to a tablespace.
- A minimum of 3 redo log groups with a minimum 1 redo log each.
- UNDO tablespace can't be used for anything but internal UNDO processing
- All undo is also written to the REDO logs for recovery purposes.

Unique Objects

- Package and Package Bodies- groups of procedures, functions, variables, cursors, etc.
- User Defined Objects types
- Nested tables and VARRAYS, aka collections
- Sequences are the default for sequence generation.
- The DUAL table: SELECT 8*6 from DUAL;

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Sessions and Processes

- User log into, no logins, (changes a bit with multitenant architecture with c## global users)
- Each session spawns a new process on the OS, (vs. threads in SQL Server)
- How has this changed with multitenant?

Security

- Built in roles and privileges
- Deeply complex row level security, encryption and isolation.
- All processes owned by a designated ORACLE user
- Advanced auditing, (revamped in release 19c)

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Interacting with Oracle Data

- Application Interface
- Development and Query Tools
  - SQL Developer
  - VS Code
  - PL/SQL Developer
  - SQL*Plus
  - SQLcl

Connection Information

- Info Required:
  - Host Name or IP Address
  - Port
  - Connection Type, (TCP, EZ Connect, TCPS, Bequeath, Wallet)
  - SID, Service Name

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Connection Strings and Connecting to Oracle

Bequeath connection as DBA:

```
>sqlplus / as sysdba
```

Standard user connection to a multitenant PDB using SQLcl:

```
>sql –s -L jdoe@//dbhost:1521/orclpdb1
```

EZConnect:

```
>sqlplus jdoe@host.com:1521:ORCL
```

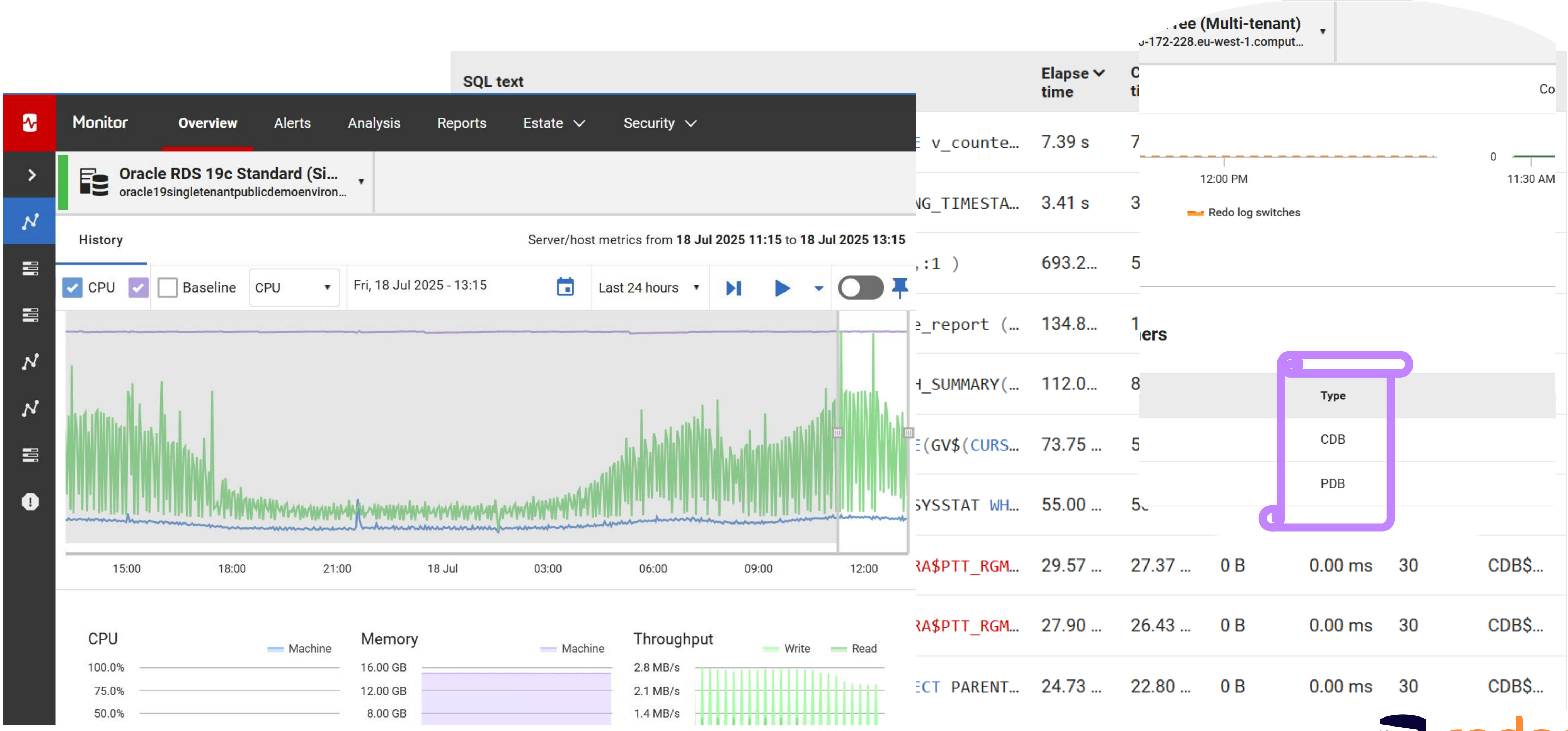
Oracle Architecture

The Oracle Evolution

Monitoring the Behemoth

- Our teams discovered the long runway of identifying the architectures that would be required.
- Built these out and how different the roadmap for Oracle was vs. other database platforms, (even SQL Server!)

Long Runway of Architectures



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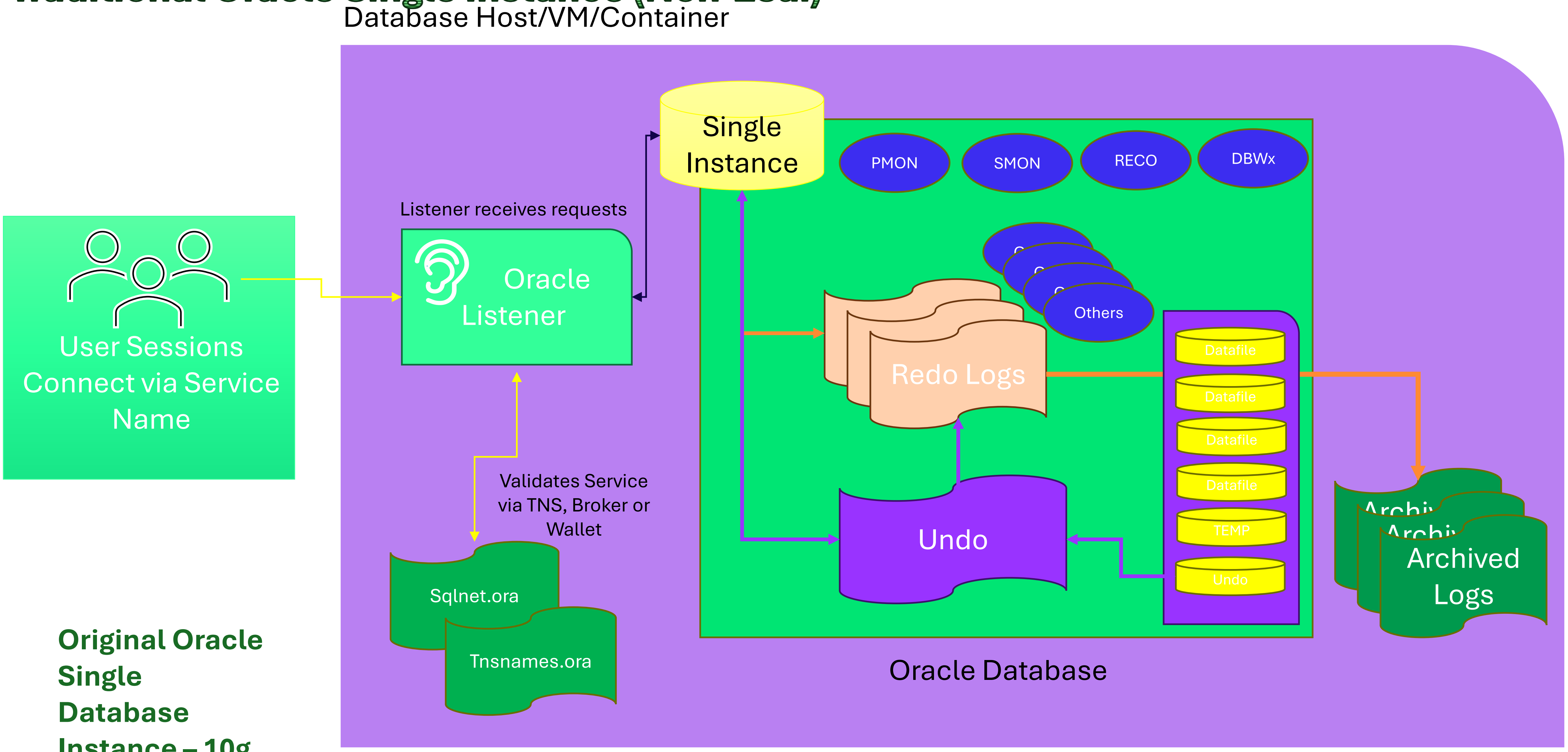
Historical Single Instance Oracle

- Owns all background processes.
- Requires more resources to run a database, pushing for larger monolith systems.
- Limits the ability to migrate to the cloud.
- More difficult backup/recovery and high availability solutions.
- No longer available in Oracle 23ai

Traditional Oracle Single Instance (Non-23ai)



Original Oracle Single Database Instance – 10g to 21c

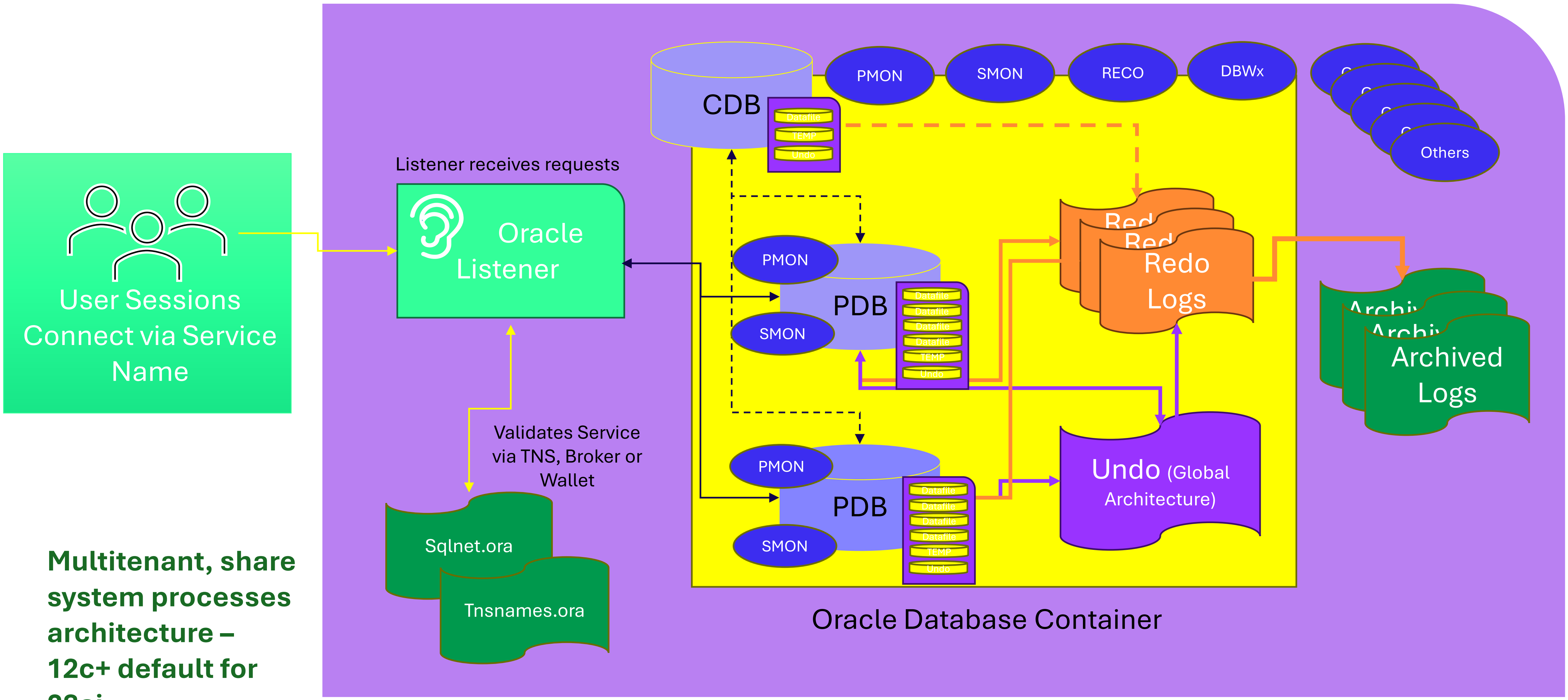
Multi Tenant is Introduced in Oracle 12c

- Container Database
  - Like Master Database in SQL Server
  - Own majority of background processes, manages “containers”.
- Pluggable databases
  - Like user databases in SQL Server
  - Simplifies management, migration and uses significantly less resources.
- Seed Database
  - Template database to ease cloning and database creation.
  - Like Model database in SQL Server

**A MULTITENANT license is not the same as MULTITENANT architecture*

High Level Oracle Multitenant(Container)

Database Host/VM/Container



Multitenant, share system processes architecture – 12c+ default for 23ai

What is RAC?

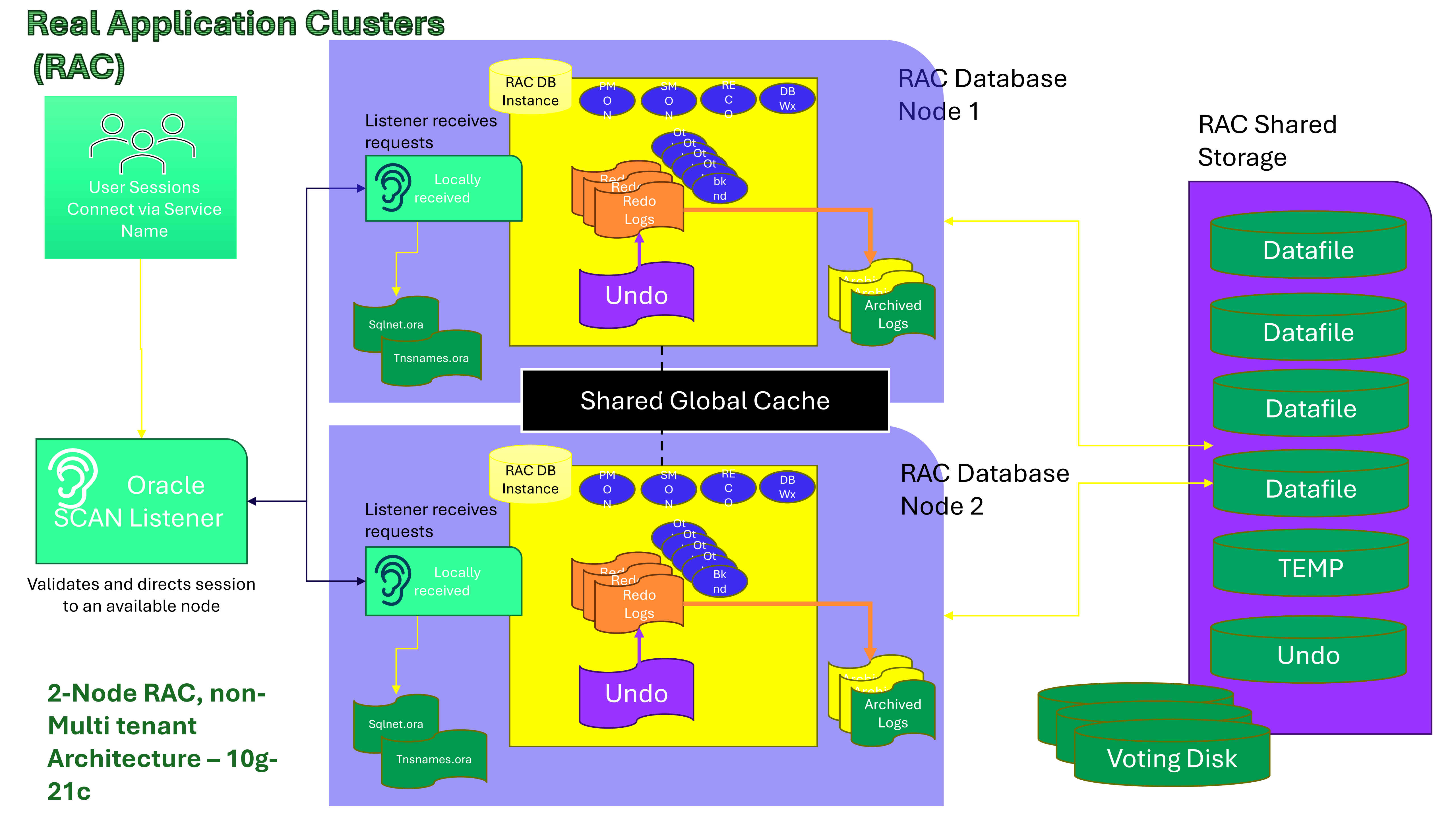
Like Windows Failover Clustering, (WFCS)

Instance resiliency and scalability, but often sold as HIGH AVAILABILITY, (respect the propaganda on this one! ☺)

Shared storage and unable to span multiple datacenters, (why it fails HA 101).

Can scale 32+ nodes, as long as database design and code can also scale!

RAC One-node offers an active/passive solution for failover in single datacenter.


Real Application Clusters (RAC)
User Sessions Connect via Service Name
Oracle SCAN Listener
Validates and directs session to an available node
2-Node RAC, non-Multi tenant Architecture – 10g-21c
RAC DB Instance
PMON
SMON
RECO
DBWx
Listener receives requests
Locally received
Redo Logs
Undo
Sqlnet.ora
Tnsnames.ora
Archived Logs
RAC Database Node 1
Shared Global Cache
RAC DB Instance
PMON
SMON
RECO
DBWx
Listener receives requests
Locally received
Redo Logs
Undo
Sqlnet.ora
Tnsnames.ora
Archived Logs
RAC Database Node 2
RAC Shared Storage
Datafile
Datafile
Datafile
Datafile
TEMP
Undo
Voting Disk

Oracle RAC vs. SQL Server Always-on

Unlike SQL Server Always-on Availability Groups, there isn't a replica of the database- single database, shared between multiple DB servers. Transactional processing is "shipped" between the nodes, (servers) to keep them in sync.

No write BETWEEN nodes on AG. In Oracle RAC, there is a share everything design.

There is a significant overhead to this, global cache, (gc waits)- CPU, memory and network, although there are some similar waits in AG on SQL Server.

AG is built for failover, where RAC is built for scalability.

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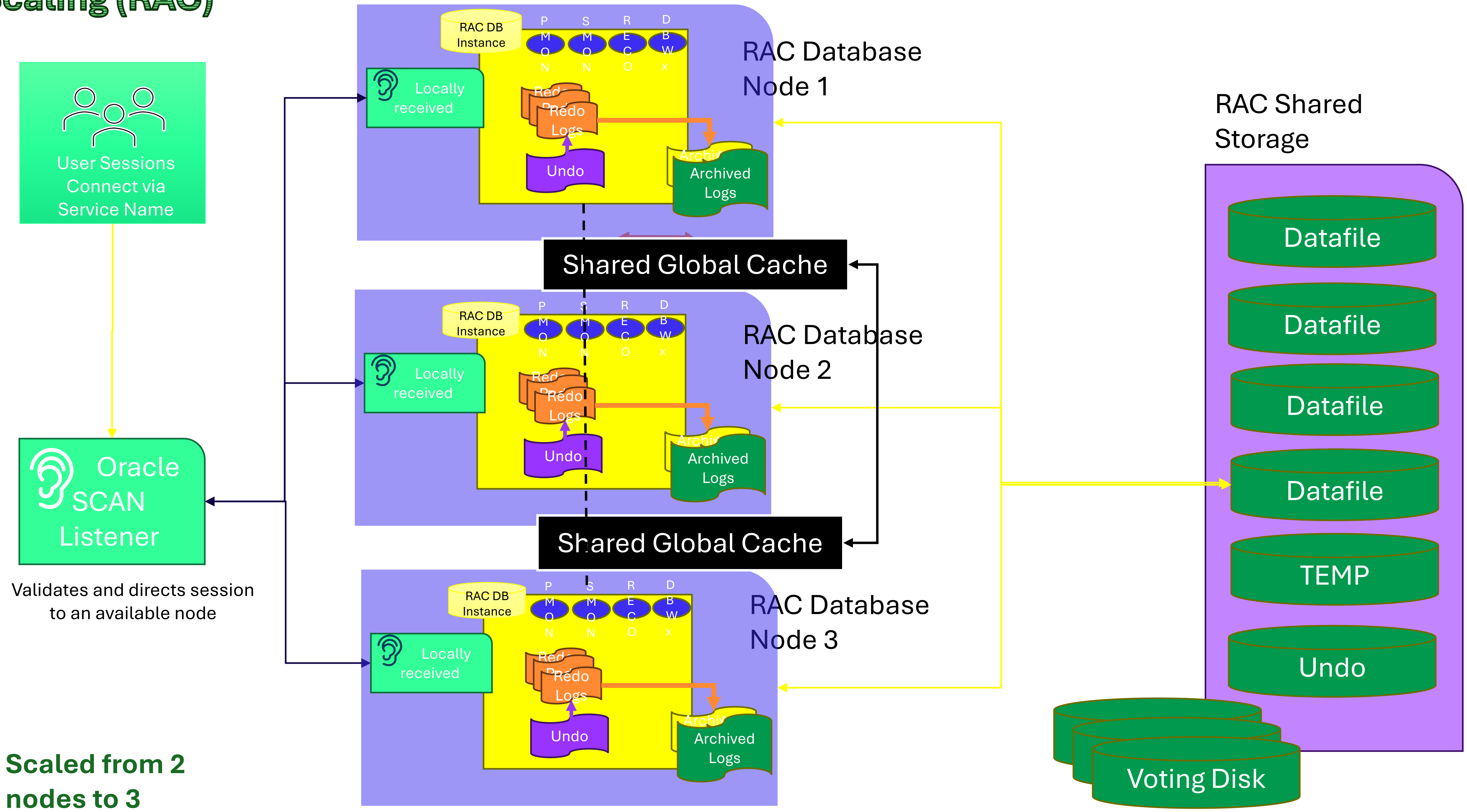
Scaling (RAC)



Scaled from 2 nodes to 3

Data Guard

Like Always-on Availability Groups in SQL Server

Active/Passive or Active/Read-only solutions

Uses terms “Primary” and “Secondaries”

Can have many secondaries and set up with failover hierarchy.

Has Fast-start failover, (FSFO) use a Far Sync host to manage data replication between long distances or multi-cloud.

Incredibly flexible and resilient.

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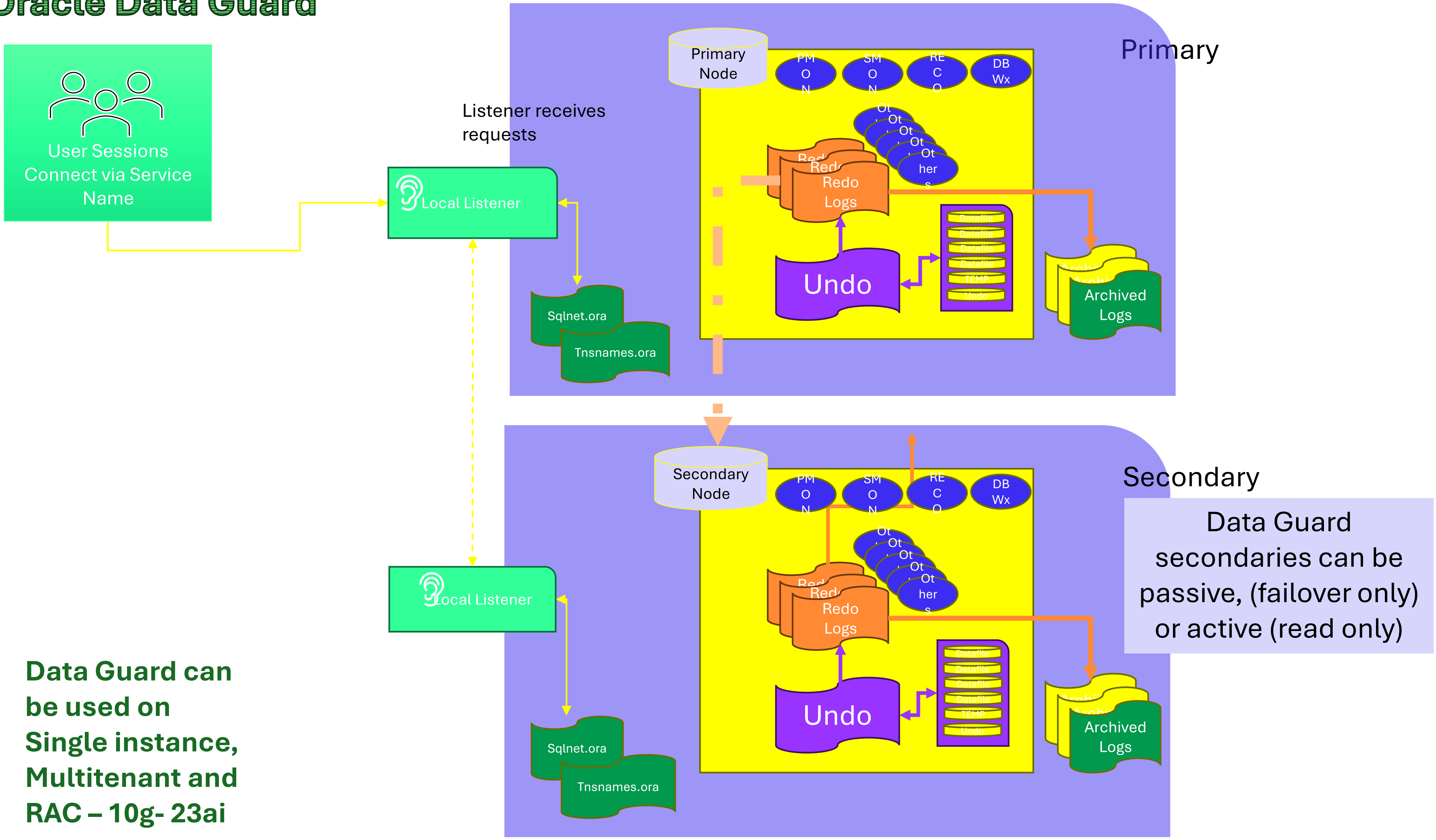
Oracle Data Guard



Data Guard can be used on Single instance, Multitenant and RAC – 10g- 23ai

The Oracle Engineered System- Exadata

What is Oracle Exadata?

An Engineered hardware and software solution specifically designed to run Oracle workloads

Consists of database and cell nodes, (servers), fast internal network, PDU, (power) in a rack configuration.

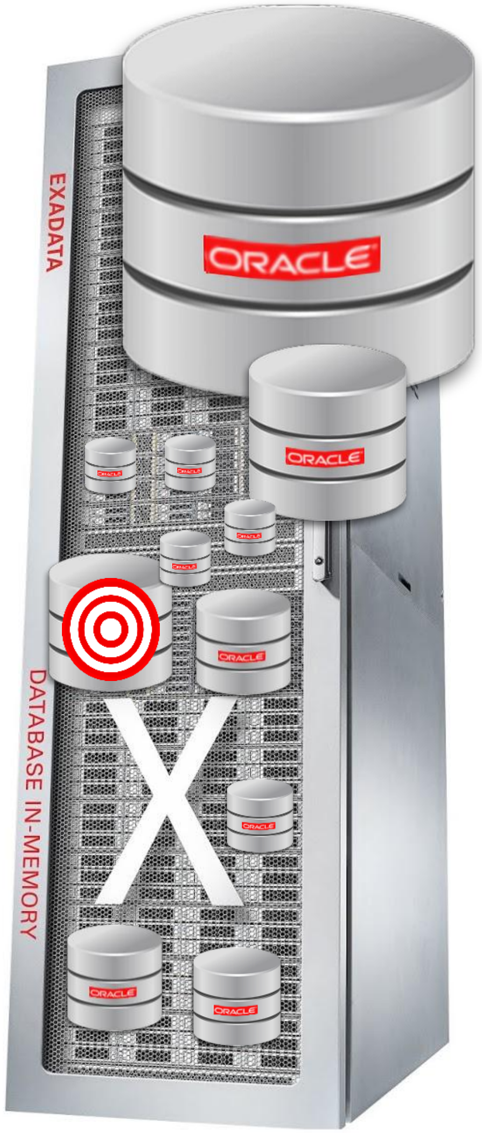
Proprietary features:

- Offloading, (using cell nodes to perform high IO processing instead of the primary database nodes.
- Storage indexes, (indexes in memory, along with advanced features.)
- Flash cache and Flash Logging for faster buffer results, elimination of log latency
- HCC, (Hybrid Columnar Compression) for faster scans of compressed data
- Secondary features depending on the version of Exadata(X5-2L through X11-M versions are available, depending on the location of OCI, Exadata@customer [on-premises] or Exadata@cloud).

Can use Oracle RAC or RAC single node, (active/passive)

Software intelligence installed to know when to use Exadata features to benefit the workload

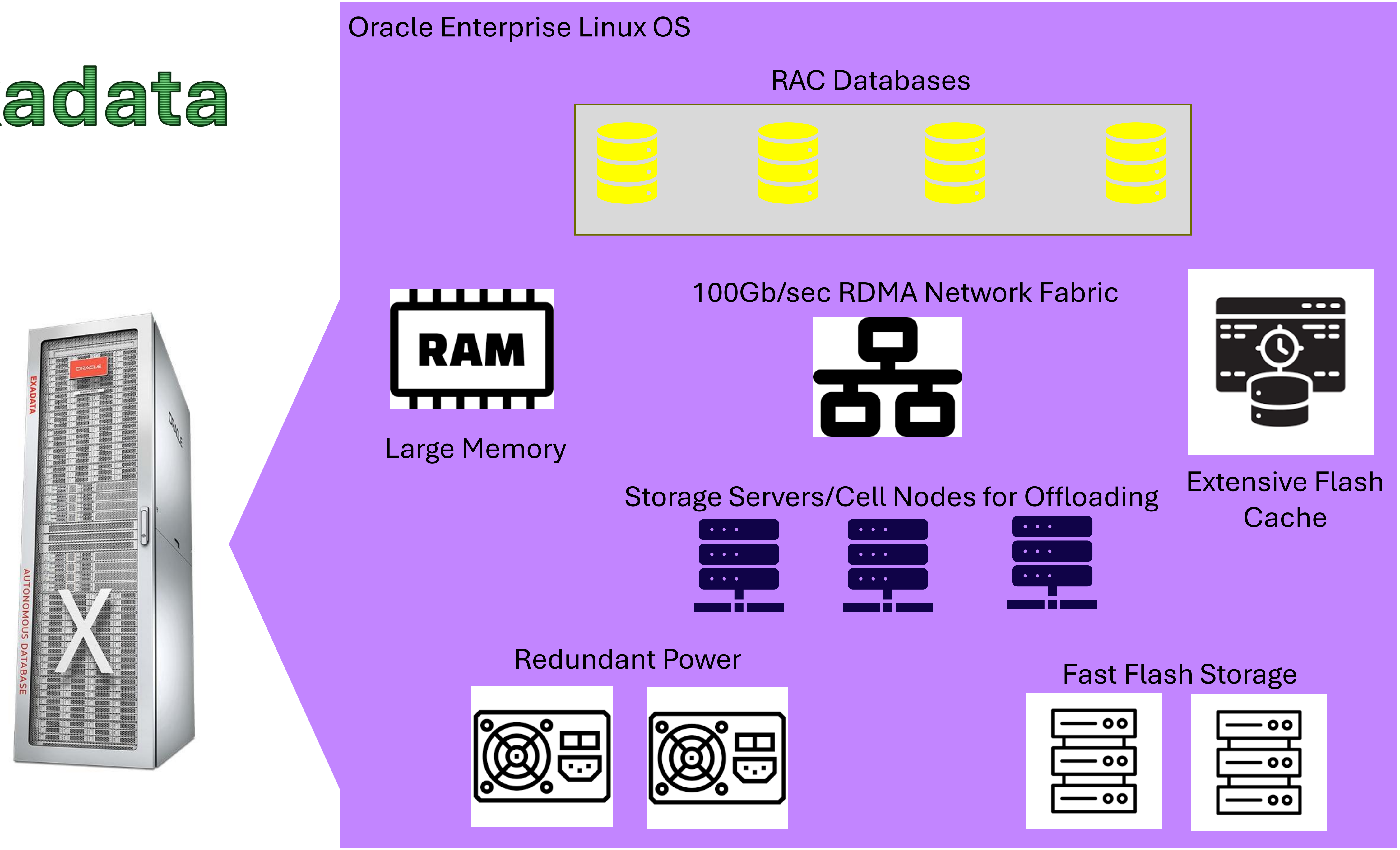
Oracle Exadata- A cloud in a box



- Purchase of Exadata was most likely made due to 1-2 large databases.
- Rest of databases were consolidated to make the most of the investment.

- Most databases heavily dependent on the infrastructure, not Exadata features.
- Can have upwards of
    - 2990 CPU cores
    - 33TB of memory
    - 21TB of RDMA memory
    - 3-7 cell nodes for offloading
    - 462TB of flash capacity
    - 4.2PB of SSD storage

Exadata


Oracle Enterprise Linux OS
RAC Databases
RAM
Large Memory
100Gb/sec RDMA Network Fabric
Extensive Flash Cache
Storage Servers/Cell Nodes for Offloading
Redundant Power
Fast Flash Storage
EXADATA
AUTONOMOUS DATABASE

Why Oracle has Lock-in

Oracle Modernization Failures

- Uses PL/SQL
- Dependent upon internal PL/SQL packages (collection of procs, functions, etc.) to limit demands on developers.
- Schema migration of objects is easy, recoding of PL/SQL is NOT.

In Summary

Oracle is...

- Much more than a database platform
- Is expensive but has lock-in around it’s PL/SQL, internal packages, instance resiliency and scalability that other platforms will have challenges in the modernization game.
- Greenfield projects rarely make sense to choose Oracle.

Oracle requires knowledgeable resources

- As it’s part of 100% of fortune 500+ companies.
- Requires deep database, infrastructure, development and command line skills.
- Has advanced architectures differing greatly from other database platforms.
- Has differing definitions of common terms.

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Thank you

Kellyn and Rene

Kellyn:

Rene:

Your feedback is important to us

Evaluate this session at:

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Take a Brain Break
Come and build some LEGO!

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